

NFV: Use Cases

Programmable Networks A.Y. 2024/25

Outline

- Introduction
- Use Case #1 - NFVlaaS
- Use Case #2 - VNFaaS
- Use Case #4 - VNF Forwarding Graphs
- Use Case #17 - Network Slicing
- Use Case #5 - Virtualisation of Mobile Core Network and IMS

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Introduction

- A **Softwarized Network** is represented by a set of Physical and Virtual Network Functions (**PNF** and **VNF**) **interconnected to provide a Network Service** to end users
- The key concept allowing the realization of a Softwarized Network is the **decoupling of Network Functions from the underlying hardware** (a.k.a. Network Function Virtualization, **NFV**)
- ETSI defined a set of **use cases** to emphasize the advantages of Softwarized Networks

“We’re certainly interested in NFV for new and expanding deployments. In order of importance, we want to bring up new services more quickly; we want reduced operational cost, and lastly capital costs.”

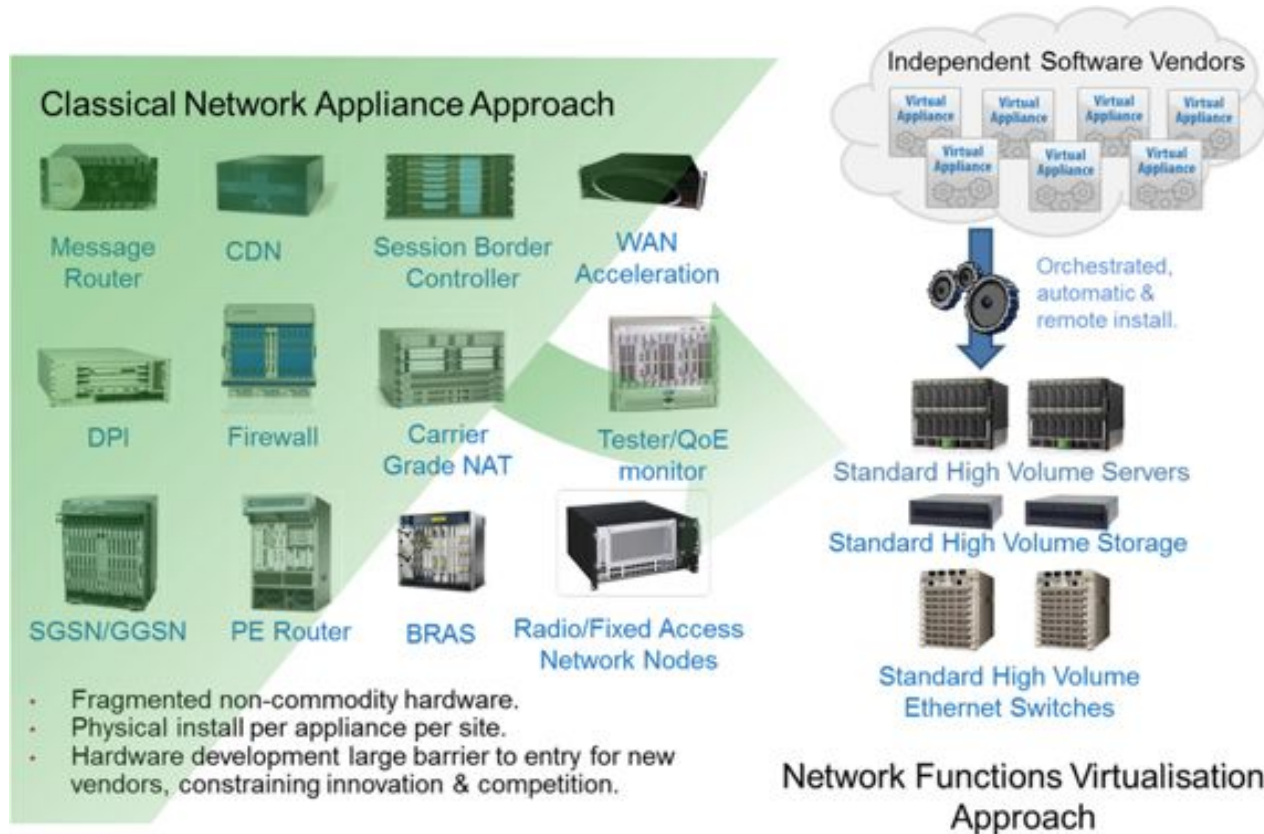
Sprint

What is Network Function Virtualisation?

«NFV aims to transform the way that network operators architect and operate networks and network services by evolving standard IT virtualisation technology to consolidate many network equipment types onto industry standard high volume servers, switches and storage»

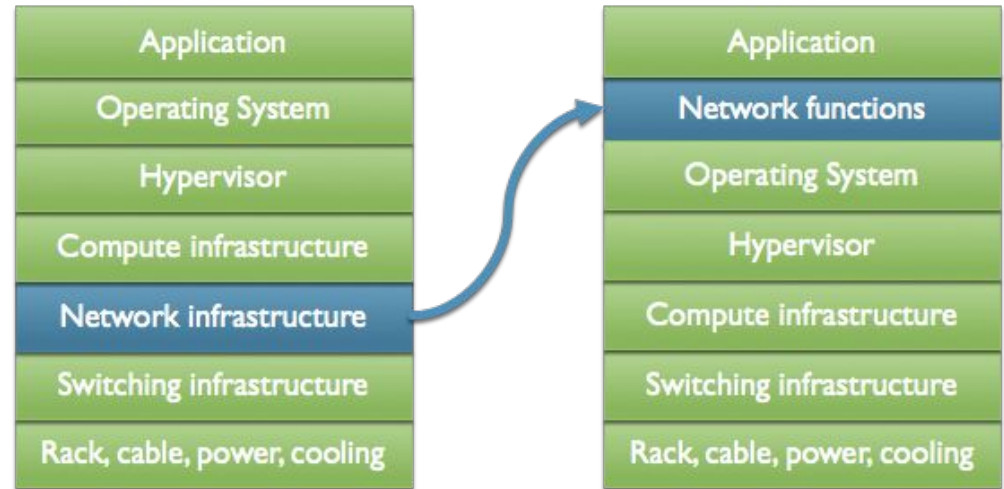
Network Functions Virtualisation – Update White Paper, available on the ETSI website

What is Network Function Virtualisation?

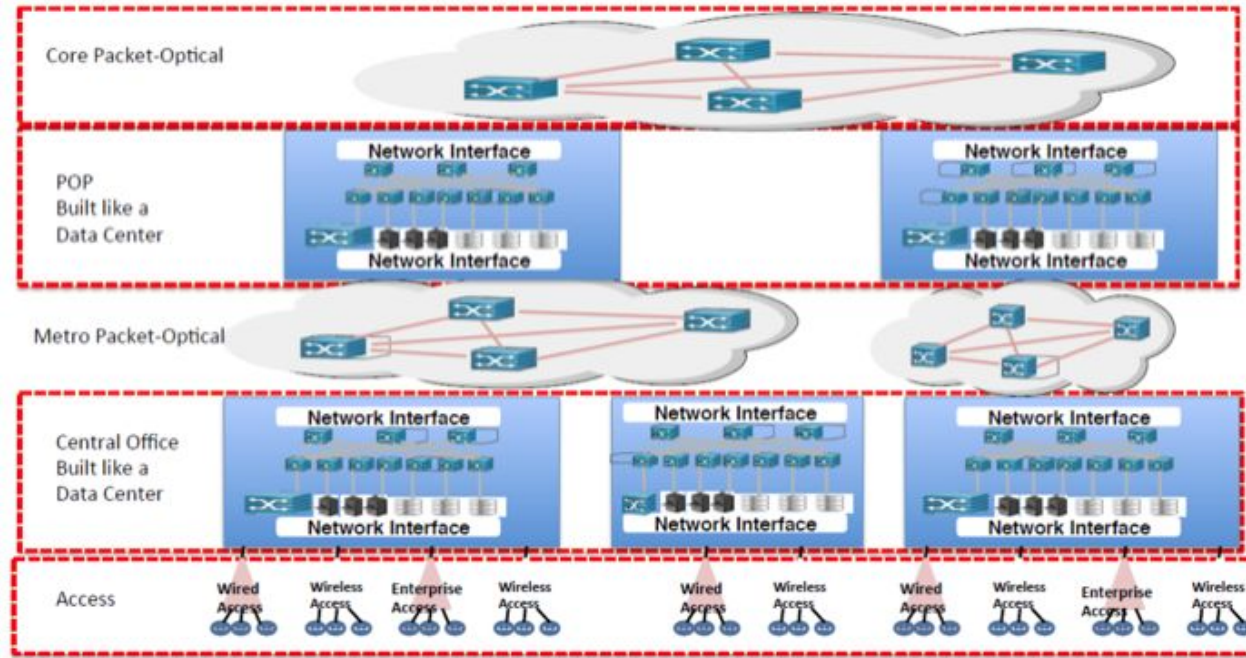


Network Function as Software

- **New Design Challenges**
 - scaling in the cloud/fog rather than optimize hardware nodes
- **New Operational Challenges**
 - security, provisioning, management, etc.



Future Network Architectures



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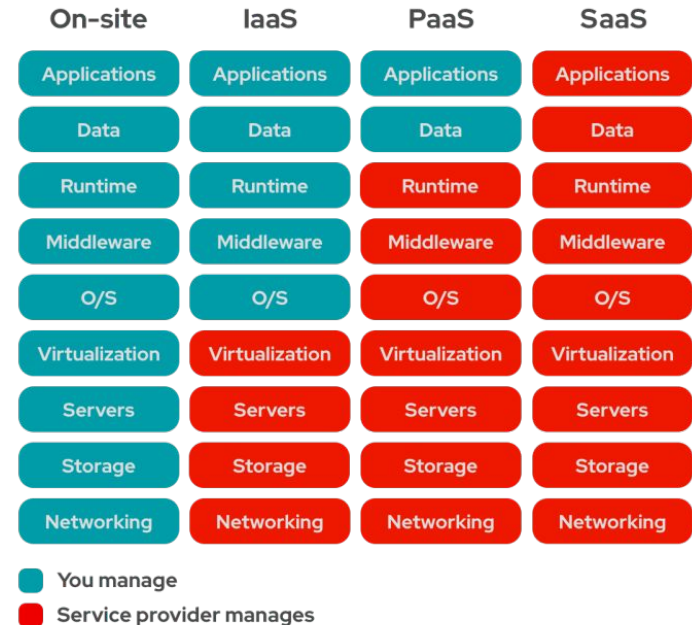
Use Case #1

Network Functions Virtualisation Infrastructure as a Service (NFVlaaS)

- Many Service Providers offer cloud computing services in addition to network services (acting as Cloud Service Providers- CSPs when doing so)
- SPs are asked to meet stringent network service performance objectives (e.g. latency, reliability) at large scale
 - few SPs can deploy physical infrastructure around the globe
 - customers demand global services
- IDEA: deploy and run Virtualized network functions inside an NFV Infrastructure provided as a service by another SP

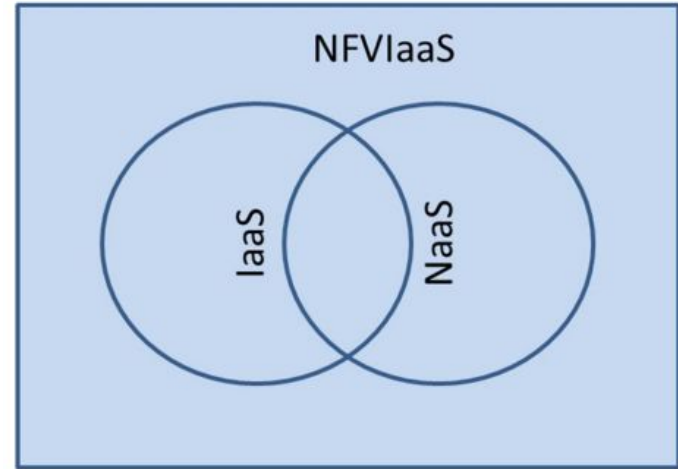
Use Case #1

- **Cloud Computing Services** are typically offered to consumers in one of three service models
 - Infrastructure as a Service (**IaaS**)
 - Platform as a Service (**PaaS**)
 - Software as a Service (**SaaS**)
- Some literature also refers to a capability to offer network connectivity services as Network as a Service (**NaaS**)

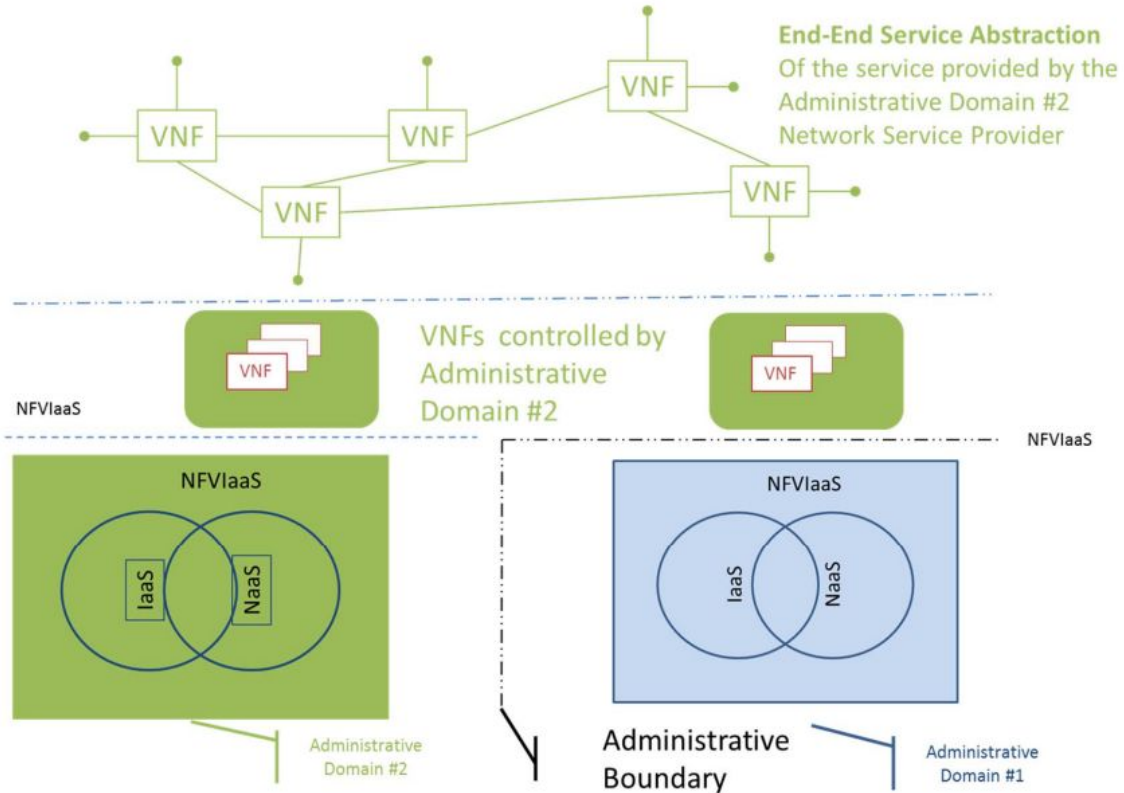


Use Case #1

- The **NFV Infrastructure** can be considered as an **environment in which Virtualized network functions can execute**
- This use case introduce the concept of **NFVI as a Service**
- The resources to be pooled are
 - physical network
 - storage
 - CPU



Use Case #1



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Use Case #2

Virtual Network Function as a Service (VNFaaS)

- Today's enterprises are deploying multiple services at the edge of branch offices
- Many enterprises find the cost of a dedicated standalone appliance per-feature prohibitive, inflexible, slow to install and difficult to maintain
- Faced with a big required investment, many enterprises are looking for outsource alternatives.
- These alternatives may include the virtualisation of the Enterprise CPE (Access Router) into the operator's network

Use Case #2

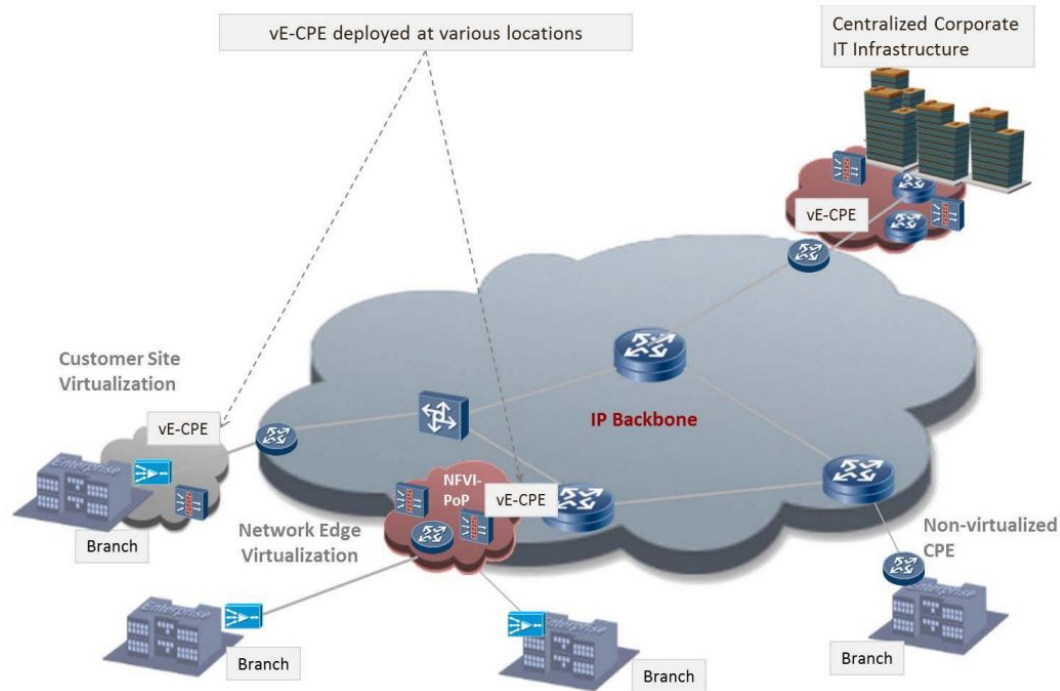


Figure 6.2.2-1: Service Provider without virtualisation of the enterprise

- Virtualization of the enterprise may include:
 - Virtualization of the CPE functions (vE-CPE) in the Service Provider cloud
 - Virtualization of the PE functions (vPE) where the virtual network services functions and core-facing PE functions can be executed in the Service Provider cloud

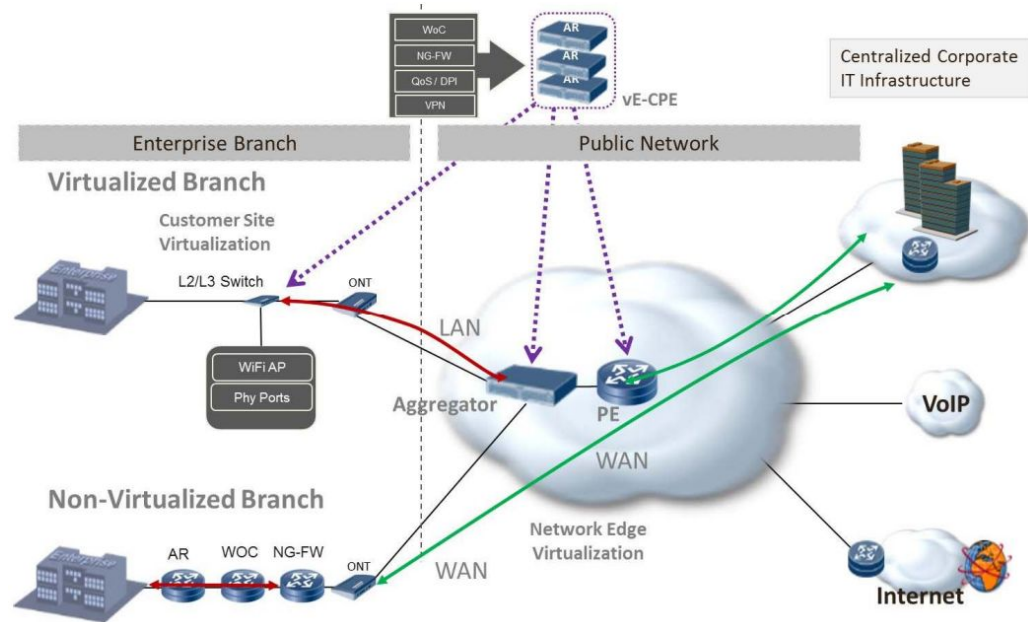
Use Case #2

Virtualization of the CPE (vE-CPE)



Use Case #2

- The enterprise local traffic is handled by a local L2 or L3 switch providing physical connectivity
- Enterprise LAN is extended to the Service Provider NFV Network located vE-CPE
- Example functionality provided by the vE-CPE includes routing, VPN termination, QoS support, DPI, NG-FW and a WOC (WAN Optimization Controller)



dotted purple lines indicate where this vE-CPE functionality may be located

Outline

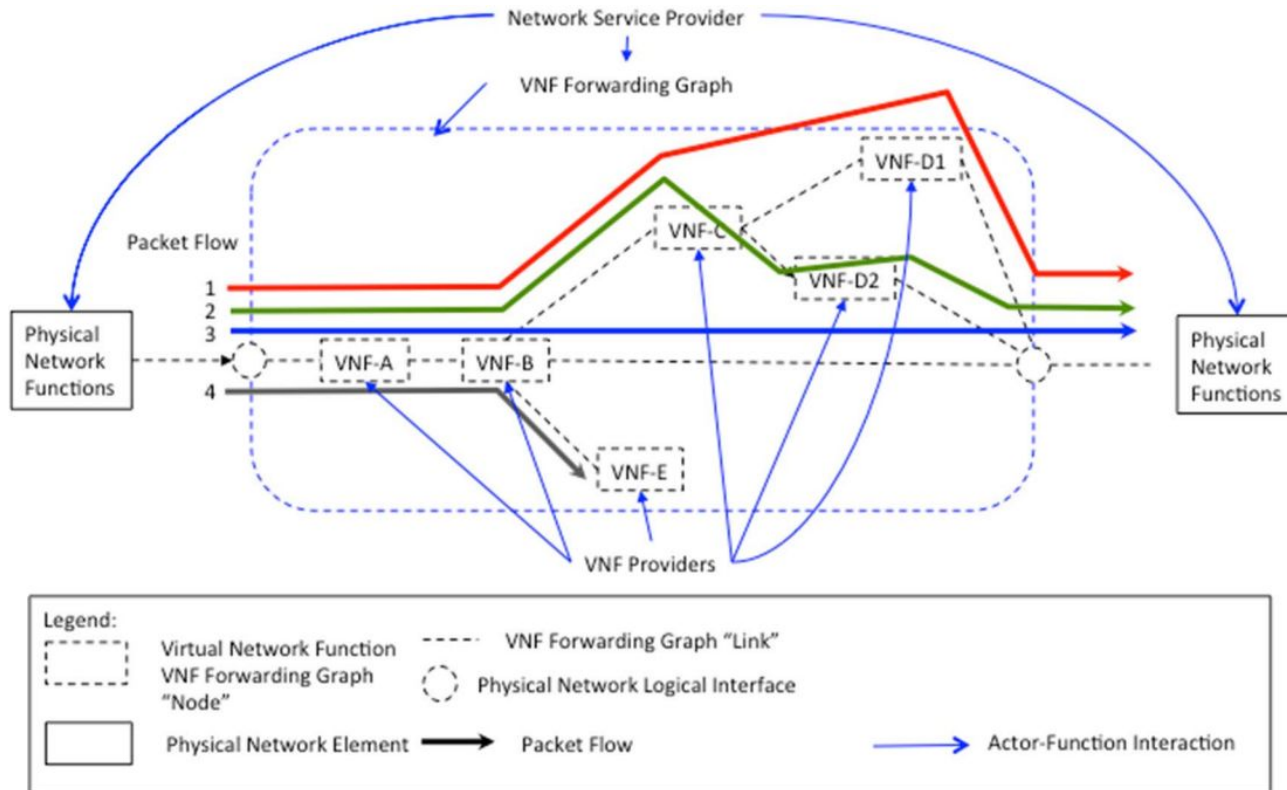
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Use Case #4

VNF Forwarding Graphs

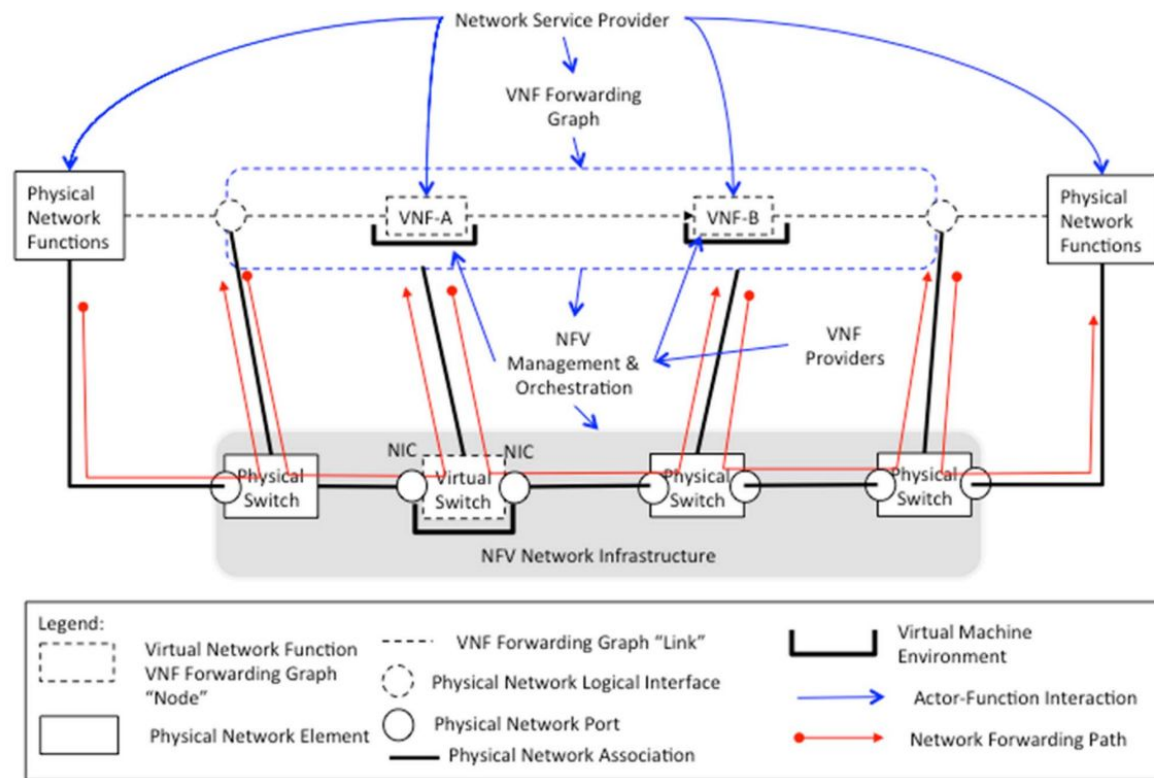
- A Virtual Network Function (VNF) Forwarding Graph defines the sequence of VNFs that packets traverse
- VNFFGs are equivalent to connecting Physical Appliances via cables
- In other words, a VNF Forwarding Graph provides the logical connectivity between virtual appliances (i.e. VNFs)
- Elements of a VNF-FG are
 - Physical Network Function
 - Physical Network Logical Interface
 - Packet Flow
 - NFV Network Infrastructure

Use Case #4



Logical View of Virtual Network Function Forwarding Graph (VNFFG)

Use Case #4



Physical View of Virtual Network Function Forwarding Graph (VNFFG)

Use Case #4

Attribute	Physical Appliance Forwarding Graph	VNF Forwarding Graph
Efficiency	Dedicated function and network capacity sized for peak load	Function and network capacity sized to current load and shareable across functions
Resiliency	Backups use specific hardware and dedicated network capacity	In some cases, backup functions can share hardware resources and network capacity in the NFV Infrastructure
Flexibility	Lengthy deployment intervals for upgrades or new features when functions are hardware based	Shorter deployment intervals for upgrades and new features since functions are software based
Complexity	Additional configuration, physical interfaces and/or support systems needed to make client-server IP/Ethernet switching implement middlebox forwarding graphs	Virtualised switching functions and/or configuration of VNFs can implement forwarding graphs in a more straightforward and efficient manner
Deployability	Deployment in another Operator's or Enterprise's network requires physical boxes, interfaces and configuration to connect to end users	Virtualised functions and switching more easily deployable in Operator's or Enterprise's network. Virtualisation of networking functions can reduce configuration complexity

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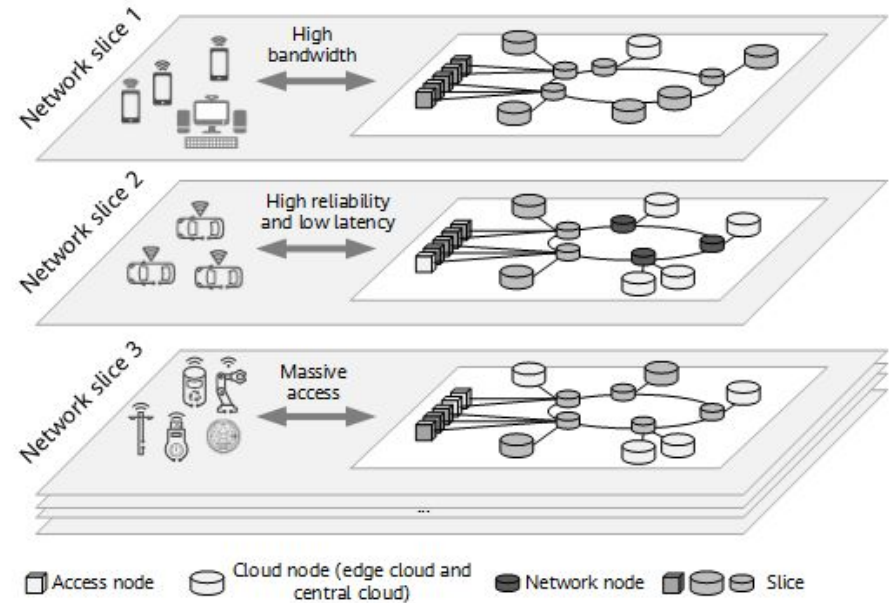
Use Case #17

Network Slicing

- The concept of Network slicing consists in **running multiple logical networks on a common physical infrastructure**
- Examples of network slices are
 - for IoT devices to connect to the network and reach back end M2M applications
 - for smartphones to connect to the same network but to reach out to VoLTE IMS server
- In both cases, the devices connect to the same service provider infrastructure but they use either a different set of virtual or physical functions, or a different path

Use Case #17

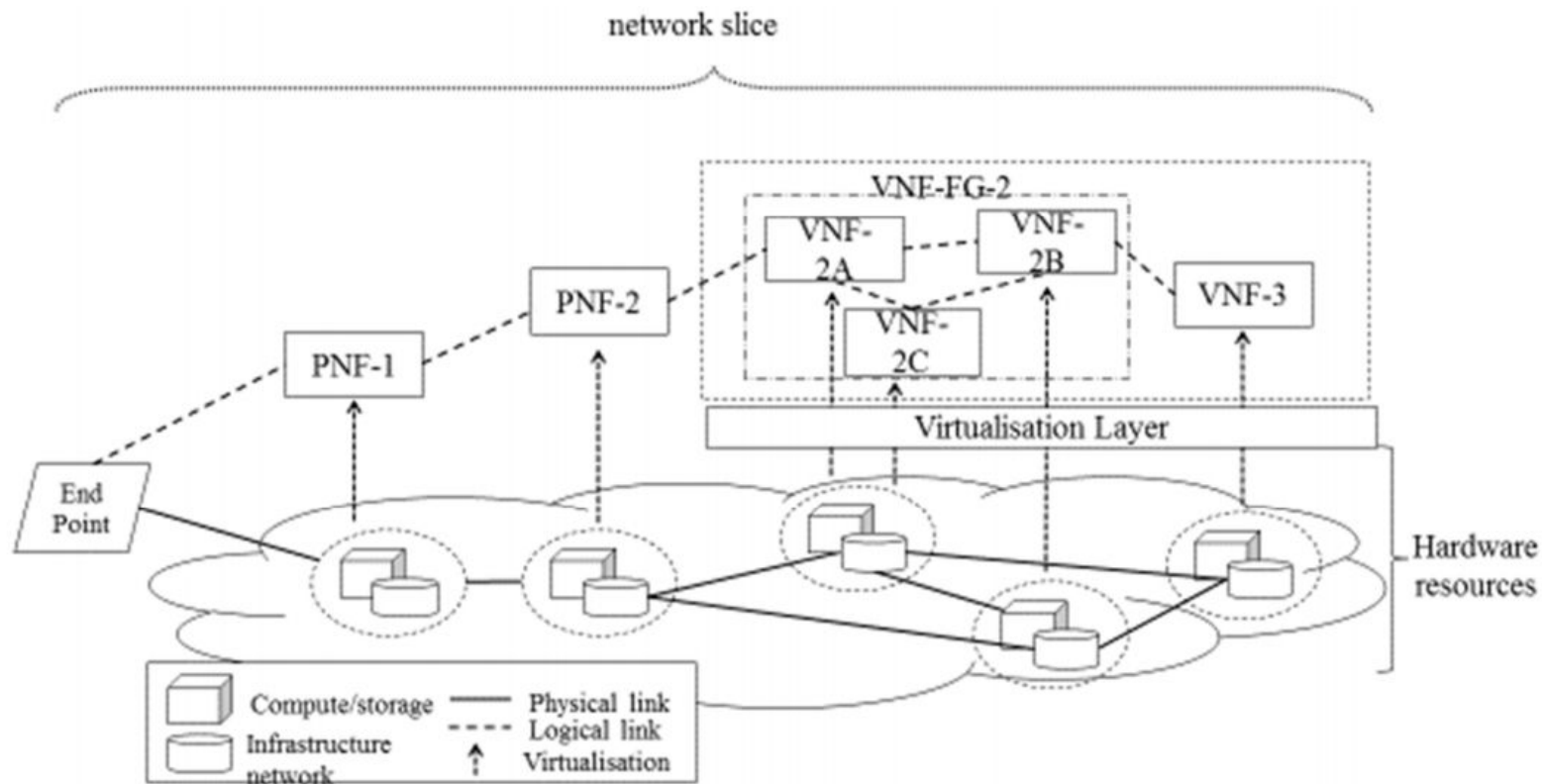
- **eMBB** (enhanced Mobile Broadband)
 - streaming of High Quality Video, fast large file transfers, etc.
- **URLLC** (ultra- reliable low latency communications)
 - applications including, industrial automation, (remote) control systems, etc.
- **MIoT** (massive IoT)
 - support of a large number and high density of IoT devices efficiently and cost effectively



Use Case #17

- The slicing idea is based on the concept that **each slice will be created to cover the demand for one or more services**:
 - providing connectivity between endpoints (terminals or network gateways)
 - processing traffic in between end-point where needed
 - providing network and Services management capabilities (OSS) and its own Real Time network and service management capabilities
 - providing support for the business administration (BSS)

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Use Case #5

Virtualization of Mobile Core Network and IMS

- Mobile networks are populated with a large variety of proprietary hardware appliances
- Possible advantages of the virtualisation of mobile core network and IMS include the following:
 - Reduced Total Cost of Ownership (TCO)
 - Improved network usage efficiency
 - Higher service availability and resiliency
 - Network topology can be dynamically reconfigured to optimize performances

Use Case #5

